Schnorr's Proof of Knowledge

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Protocols for Proof of knowledge

1 Schnorr's Dlog Proof of Knowledge

1.1 Protocol

- g $\rightarrow$ Group generator
- p = ord($\mathfrak{G}$)

Prover (w,$x = g^w$) | **Verifier**(x = g^w)

- r $\xleftarrow{R} Z_p$
- a = g^r

$\longrightarrow \xrightarrow{a} \longrightarrow$

c $\xleftarrow{R} Z_p$

$\longleftarrow \xleftarrow{c} \longleftarrow$

z = r + c.w $\longrightarrow \xrightarrow{z} \longrightarrow$ Check g^z ?= a.x^c

1.2 Properties

1.2.1 Knowledge Extractibility/ SOundness

The extractor runs the protocol twice while a remains the same in both runs.

Prover*(malicious) (w,$x = g^w$) | **Extractor** (x = g^w)

$\longrightarrow \xrightarrow{a} \longrightarrow$

$\longleftarrow \xleftarrow{c*} \longleftarrow$

$\longrightarrow \xrightarrow{z*} \longrightarrow$

- $\frac{g^z}{g^{z*}} = x^{c-c*} = g^w c - c*$
- w = $\frac{z-z*}{c-c*}$

1.2.2 Zero knwoledge HVZK

HVZK because if V* chooses c adaptively (not randomly), then Simulator is NOT Indistinguishable.

Simulator (w,$x = g^w$) | **Verifier*(mal)** (x = g^w)

$$\longrightarrow \xrightarrow{a} \longrightarrow$$
$$\longleftarrow \xleftarrow{c} \longleftarrow$$
$$\longrightarrow \xrightarrow{z} \longrightarrow$$

- z $\xleftarrow{R} Z_p$
- c $\xleftarrow{R} Z_p$
- a = $\frac{g^z}{x^c}$

1.2.3 Completeness

It is trivial.